

THE FORGE

FIELD DOSSIER ·
BALANCE SANDBOX

DESIGN SAMPLE · SEED

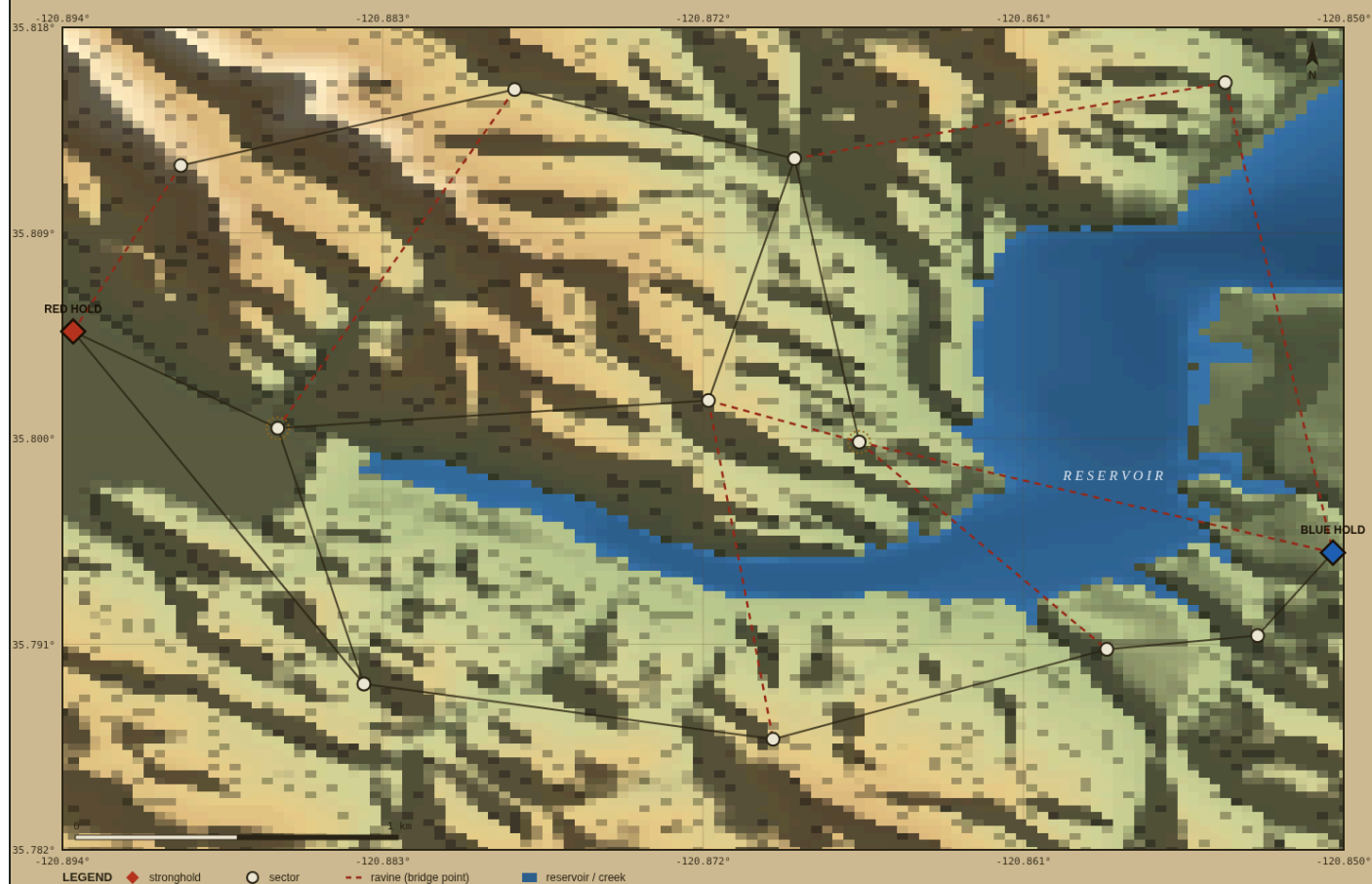
Game Systems & Mechanics Designer
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An **in-silico** playtest for one system of The Forge — territory, the bridge momentum layer, player classes, and a siege endgame. You can't cheaply playtest a 500-person, 90-minute field game, so this models one system and runs thousands of matches headless to read the balance *before* anyone books a venue. One rules engine drives the live match **and** the sims — a designer/GM pre-flight rig. It is the deliverable the role names: a **living design document**, **balance logic**, and **playtest frameworks**, not just code.

I. THE FIELD

PLATE I · FIELD SURVEY

SRTM DEM · Z13



THE FORGE — FIELD SURVEY

Lake San Antonio, Bradley CA · 35.80°N 120.88°W · SRTM DEM · 15 sectors / 8 chokepoints

Lake San Antonio, Bradley CA — a real Lightning-in-a-Bottle venue. The reservoir & creeks form the bridge-gated ravine; control points snap to dry ground; travel-time follows real slope. Chokepoints are read off the terrain, not drawn by hand.

II. FIELD NOTES — WHAT THE SANDBOX FOUND

01 SELF-PLAY FOUND AN EXPLOIT THE HEURISTICS DIDN'T

A policy trained from a blank genome reaches **~100%** against the four hand-written bots — it surfaced a hole a scripted pool would have called "balanced." RL here is a *better playtester*, not flashy AI. (CEM, no GPU; gym-microrts/PP0 skeleton ships for the scaled build.)

02 TERRAIN FLIPS BALANCE

The same ruleset that's fine on a symmetric board goes **aggressor-dominant** (~75%) on real Lake San Antonio terrain, and the venue runs side-biased. Balance is *per-venue*, not one-and-done — that's the recurring service.

03 POLICIES DON'T FULLY TRANSFER · TURLING IS NON-VIABLE

The board-trained policy slips ~85% → ~70s% on real terrain (a transfer gap), and pure turtling can't cross the ravine. Both surface as design calls, each with the levers to change them.

III. EVIDENCE

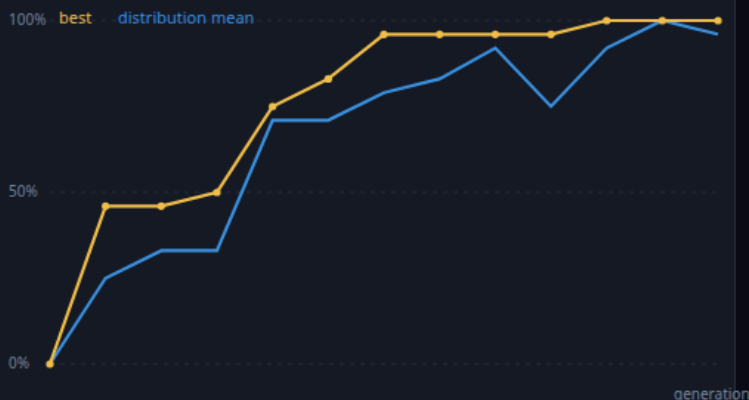
PLATE II · SELF-PLAY LEARNING CURVE

CEM · 12 GEN

The Forge — self-play playtester (CEM), learning curve

From a **blank genome** (gen 0 = naive play) to **100% win-rate** over 12 generations — learned from scratch, no GPU, dependency-free (CEM). PPO/gymnasium skeleton: `rl/forgenv.py`.

Win-rate vs the heuristic pool (expander · aggressor · bridger · turtle)



What it discovered (learned weights)

proximity	1.44
neutral	1.50
enemyOwned	-0.43
ownOwned	-2.01
enemyStronghold	0.01
siegeTowerP4	2.84
contested	0.52
homeProximity	-0.95
isTower	0.16

Positive = the policy seeks it. The win comes from weighting the **Phase-4 siege tower** and forward pressure, and *down*-weighting sitting on owned ground — i.e. it taught itself to rush the siege, the same exploit the Monte-Carlo flags.

Blank genome → ~100% win-rate vs the heuristic pool, with the weights it discovered (it taught itself to rush the Phase-4 siege tower).

PLATE III · BALANCE READOUT (REAL FIELD)

600 MATCHES · SEED



Findings, win-rate by strategy, faction fairness, and the matchup matrix – generated on the Lake San Antonio field.

IV. INSTRUMENT

LEVERS 10 dials, each with an honesty ledger (intent / trade-off / failure).	PLAYTEST Monte Carlo over all matchups: fairness, dominance, snowball, dead-time, siege rate.
MECHANICS presence-by-pulse capture, classes, the 90-min phase arc, walls + battering + delay towers, fog.	STACK ES-module JS (no build, on Pages) · Python+Pillow for the DEM · seeded (<code>node test/smoke.mjs</code>).

V. THE KIT

FIELD MANUAL the full rules as a tabletop-style rulebook — classes as stat blocks, the 90-min arc, every lever (attached / in repo).	LIVE DEMO play it, drag the levers, run the playtest — alyasska.github.io/seed-artifact
DESIGN + BALANCE DOCS game design, balance report, real-world deployment notes — in <code>docs/</code> .	REPRODUCIBLE <code>node test/smoke.mjs</code> · <code>node rl/selfplay.mjs</code> · <code>python tools/fetch_dem.py</code> .

END VISION – the pre-flight for The Forge: tune economy, class weights, and phase timings **per venue** in sim → ship a near-balanced rulebook → spend the few expensive on-field pilots validating how real humans behave, not discovering that turtling is broken with 500 people on a field.

SCOPE · one system, finished small. Classes / intel / parley designed in full, partly wired; bots map the structural envelope, not the feel. **LIVE** alyasska.github.io/seed-artifact **CODE** github.com/Alyasska/seed-artifact